

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

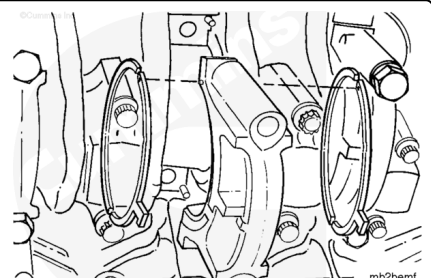


Cummins Inc., recommends replacing the thrust bearings when the main bearings are replaced.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)

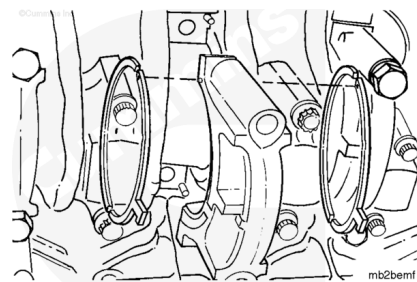
Remove

The Number 4 main bearing cap has thrust bearings. The cap is located with two dowel pins.



Remove the thrust bearings from the Number 4 main bearing cap.

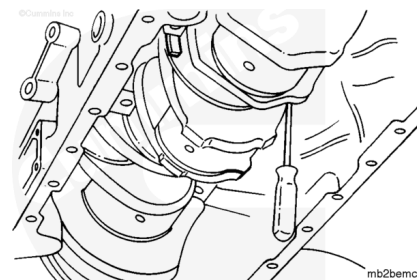
Mark the thrust bearings in the oil passage groove as being front or rear bearings.



⚠ CAUTION ⚠

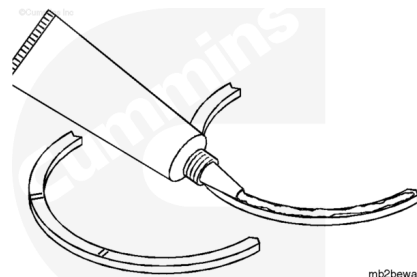
Do not damage the crankshaft when removing the upper thrust bearings.

Use a blunt tool to remove the upper thrust bearings.



Install

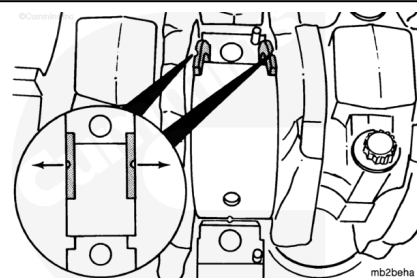
Use Lubriplate™ 105, Part Number 3163086, Part Number 3163087 or equivalent, to coat the upper thrust bearings.



Push the crankshaft toward the rear of the engine to install the rear thrust bearing, and to the front of the engine to install the front thrust bearing.

Install the upper thrust bearings in the number four main bearing saddle.

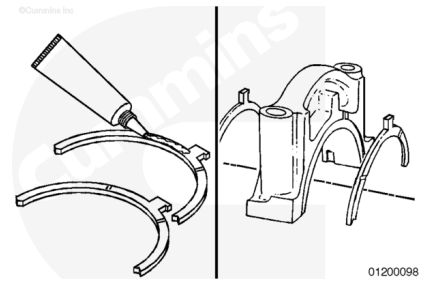
The grooves **must** be toward the crankshaft.



Use Lubriplate™ 105, Part Number 3163086, Part Number 3163087 or equivalent, to coat the lower thrust bearings.

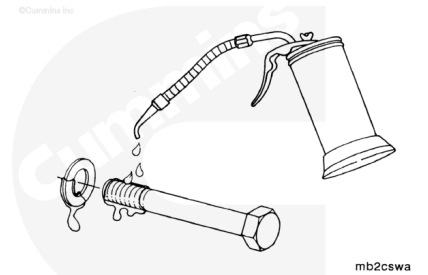
Install the lower thrust bearings in the number four main bearing cap as shown.

The grooves of the thrust bearing **must** be toward the crankshaft.



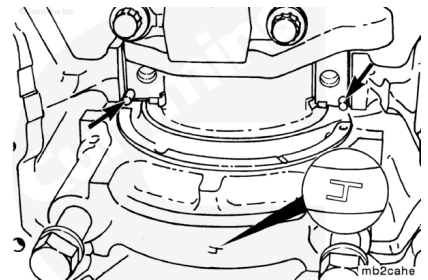
Use clean 15W-40 engine oil to coat the capscrew threads and on both sides of the washers.

Drain the excess oil from the capscrews before installing them in the cylinder block.



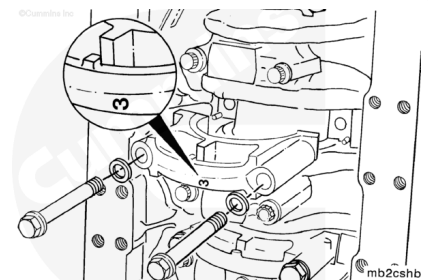
⚠ CAUTION ⚠

To avoid the possibility of damage to the cylinder block and main bearing cap, the number four main bearing cap must be aligned with the dowel pins in the bearing saddle when the capscrews are tightened.



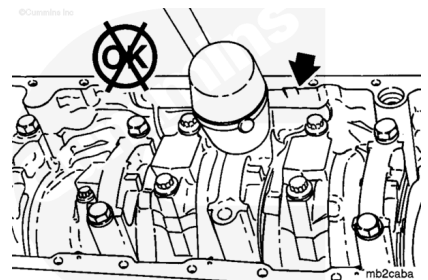
The main bearing caps are numbered one through seven from front to rear in the cylinder block. The caps **must** be installed so the number on the cap matches the bearing saddle in the block. The lock tangs in the main bearing saddle and bearing cap **must** be on the same side.

Install the main bearing caps.



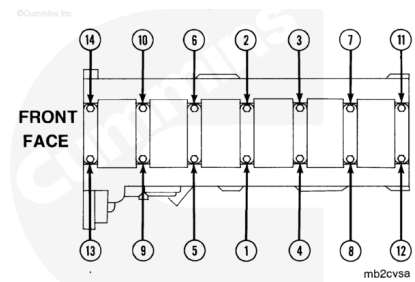
⚠ CAUTION ⚠

To avoid the possibility of damage to the main bearing cap and bearings, do not hit the main bearing cap with a hammer.



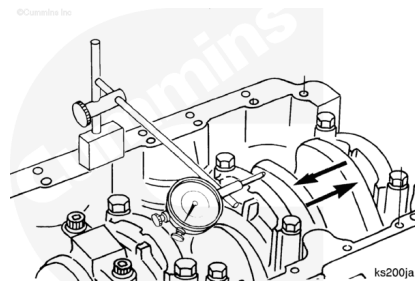
Make certain that the proper main cap capscrew torque procedure is used when torquing main bearings. The use of the improper torque procedure will damage the engine. Refer

to Procedure 001-006 in Section 1.
(/qs3/pubsys2/xml/en/procedures/101/101-001-006-tr.html)



Use a dial indicator to measure the crankshaft end clearance.

Crankshaft End Clearance		
mm		in
0.10	MIN	0.004
0.56	MAX	0.022



If the end clearance is **not** within the specifications, inspect the main bearing cap and thrust bearing surfaces in the cylinder block.

Finishing Steps

- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html)
- Operate the engine to normal operating temperature and check for leaks.

Note : The engine **must** have adequate oil pressure within 15 seconds after starting. If the warning light indicating low oil pressure has **not** gone out or there is no oil pressure indicated on a gauge within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level is in the oil pan.



